

Mathematical Derivation

Topic

Dynamic modeling of Conveyor Belt with variable load

Step 1:

System Definition

Let: m_b = mass of belt

$m_L(t)$ = time varying load

$v(t)$ = belt velocity

Total mass = $m_b + m_L(t)$

Step 2:

Identify Forces

From your free body diagram:

Driving Force: $F_{drive} = \frac{T(t)}{r}$

Friction force (Coulumb friction due to load)

$$F_{friction} = \mu m_L(t)g$$

Viscous Damping

$$F_{damping} = cv(t)$$

Step 3:

Apply Newton's second law

$$\text{Net force} = m_{total} \cdot \frac{dv}{dt}$$

$$F_{net} = F_{drive} - F_{friction} - F_{damping}$$

Step 4:

Substitute Forces

$$(m_b + m_L(t)) \frac{dv}{dt} = \frac{T(t)}{r}$$

Step 5:

Final equation (differential):

$$\frac{dv}{dt} = \frac{1}{m_b + m_L(t)} \left[\frac{T(t)}{r} - \mu m_L(t)g - cv \right]$$

Step 6: state-space Representation:

$$\text{let } x = v(t)$$

Then:

$$\dot{x} = \frac{1}{m_b + m_z(t)} \left[\frac{\tau(t)}{r} - \mu m_z(t)g - cx \right]$$

Step 7: Position Equation (optional)

$$\frac{dx}{dt} = v(t)$$

Step 8: Special case (steady state)

At steady state:

$$\frac{dv}{dt} = 0$$

$$\therefore \frac{\tau}{r} = \mu m_z g + c v_{ss}$$

$$v_{ss} = \frac{1}{c} \left(\frac{\tau}{r} - \mu m_z g \right)$$